



Example: Fixed-Increment Learning Algorithm (FILA) - training a Perceptron to solve *Boolean AND* problem (batch mode).

x_1	x_2	x_1 <i>AND</i> x_2
0	0	0
0	1	0
1	0	0
1	1	1

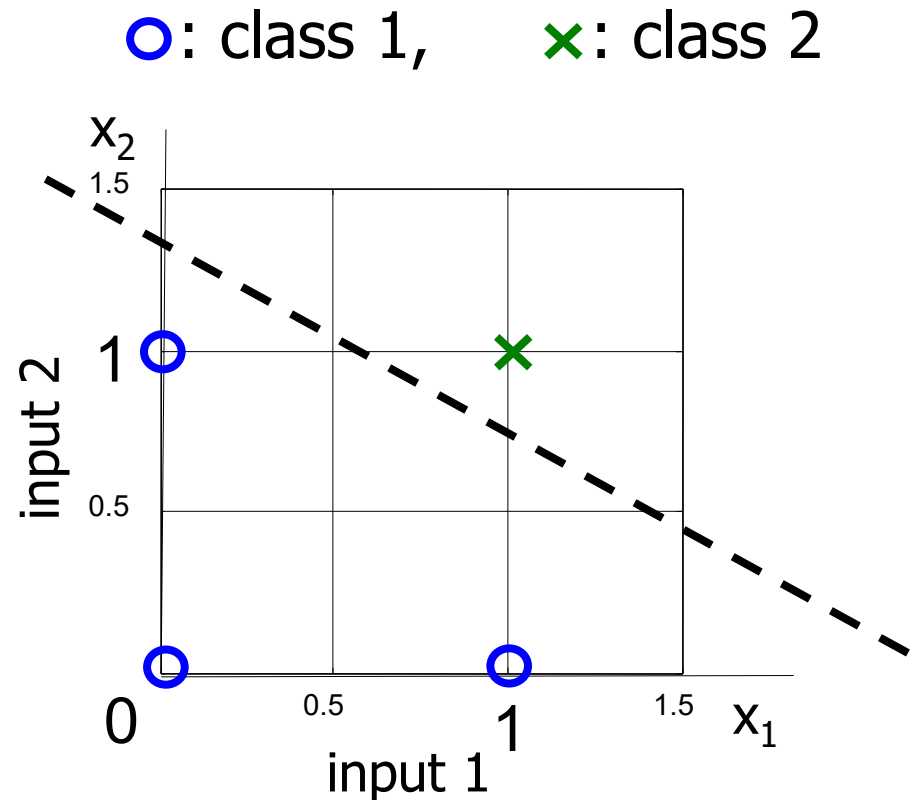


Fig. 1. The *Boolean AND* problem (linear).

Initial weight vector $\mathbf{W}^0 = (-0.2, 0.3, -0.2)$.

Transfer function - *Heaviside step* function.

Learning rate $\eta = 0.35$.

$\mathbf{X}$ - input vector. $\mathbf{S}$ - misclassified patterns vector.

$\mathbf{X}$			$\mathbf{W}$			$\bar{x} = \mathbf{X} \cdot \mathbf{W}$	$o(\bar{x})$	$\mathbf{t}$	$\mathbf{S}$		
x_0	x_1	x_2	w_0	w_1	w_2	(net)	(output)	(target)			
1	0	0	-0.2	0.3	-0.2	-0.2	0	0	0	0	0
1	1	0	-0.2	0.3	-0.2	0.1	1 	0	-1	-1	0
1	0	1	-0.2	0.3	-0.2	-0.4	0	0	0	0	0
1	1	1	-0.2	0.3	-0.2	-0.1	0 	1	1	1	1

$$\mathbf{S} = \sum_{i=1}^4 S_i = (0, 0, 1). \quad \mathbf{S} \text{ is not } (0,0,0), \text{ then:}$$

$$\Delta \mathbf{W} = \eta \mathbf{S} = 0.35 (0, 0, 1) = (0, 0, 0.35)$$

$$\mathbf{W}^{new} = \mathbf{W}^{old} + \Delta \mathbf{W} = (-0.2, 0.3, -0.2) + (0, 0, 0.35) = (-0.2, 0.3, 0.15)$$

End of one *epoch*. Because $\mathbf{S}$ is not zero, repeat, starting with $\mathbf{S} = (0, 0, 0)$ and $\mathbf{W}^1 = (-0.2, 0.3, 0.15)$.